

Demonstration of Proposed Features

This document details the live demonstration scenarios, feature walkthroughs, and anticipated Q&A; for the OptiCrop demo day.

Demo Agenda

#	Segment	Duration	Presenter
1	Problem Context	1 min	Team Lead
2	Solution Overview (slides)	1 min	ML Lead
3	Live Crop Prediction Demo 1	2 min	Frontend Dev
4	Live Crop Prediction Demo 2	2 min	Backend Dev
5	Feature Walkthrough	2 min	All
6	Model Accuracy Metrics	1 min	ML Lead
7	Q&A;	1 min	All

Demo Scenario 1 Rice Recommendation

Field	Details
Input Values	N=90, P=42, K=43, Temperature=20.9, Humidity=82, pH=6.5, Rainfall=202
Expected Output	Crop: Rice Confidence: ~99%
Talking Point	These values match the wet, nutrient-rich conditions rice thrives in. The model correctly identifies this in under 200 ms.
Visual	Show the input form being filled, then the result card appearing instantly

Demo Scenario 2 Coffee Recommendation

Field	Details
Input Values	N=19, P=18, K=27, Temperature=24.1, Humidity=68, pH=6.8, Rainfall=141
Expected Output	Crop: Coffee Confidence: ~98%
Talking Point	Coffee needs specific altitude-influenced temperatures and moderate rainfall. The model has learned these subtle distinctions from 2,200 training samples.
Visual	Contrast with rice inputs show how different soil/climate = different crop

Feature Demonstrations

Feature 1 Input Validation

Enter pH = 15 (out of range) client-side JS warning appears before submit. Clear, immediate feedback prevents server round-trip.

Feature 2 Confidence Score

Show that borderline inputs (e.g., between chickpea and lentil zones) return lower confidence scores (70-80%), giving the farmer honest uncertainty information.

Feature 3 Top-3 Alternatives

Highlight the second and third crop alternatives shown below the main recommendation. Useful when primary crop seeds are unavailable locally.

Feature 4 Mobile Responsiveness

Open DevTools toggle device toolbar select iPhone 12 show the form and result render correctly on a mobile viewport.

Anticipated Q&A;

Question	Answer
Why Random Forest?	Highest accuracy (99.09%) among 5 tested algorithms; handles feature interactions and non-linear boundaries well
What if soil test is unavailable?	User can estimate from government Soil Health Card; we show typical ranges in tooltips for each parameter
Can it handle new crops?	Retraining with expanded dataset is straightforward; Phase 2 roadmap includes 40+ crop classes
Security concerns?	No database, all inference is local, inputs are strictly validated no injection risk